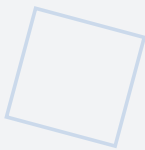


# Amagi-NIST

## V1.1

NIST AI RMF Profile for AMAGI

VERSION 1.1 — JUNE 2026



Profile mapping AMAGI architectural invariants to NIST AI Risk Management Framework controls.  
Demonstrates how hardware-enforced safety satisfies AI RMF requirements at the highest assurance level.  
Normatively references: Charter v1.2, S-APL v2.0, Framework v2.0, NIPU v1.1, MEM v1.1, Open Specification v1.1, Master Parameter Sheet v2.0.

Alexey Mikhailovich Burlai

AGIAM Technologies Ltd (In Formation)

ORCID: 0009-0001-4679-5967

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License: S-APL

# AMAGI NIST AI RMF Profile

Version 1.1 — Cross-Sectoral Target Profile

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A Subcategory-Level Target Profile Mapping the AMAGI Architectural Class  
to the NIST AI Risk Management Framework 1.0 (AI 100-1)

Alexey Mikhailovich Burlai

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S-APL v2.0 Licensed — Strict Architectural Principle License

Normatively references: Framework v2.0, NIPU v1.1, MEM v1.1, Open Specification v1.1

Supersedes: NIST AI RMF Profile v1.0 (DOI: 10.5281/zenodo.17849725)

## 1. Introduction

This document provides a NIST AI Risk Management Framework (AI RMF 1.0) cross-sectoral target profile for the AMAGI Architectural Class, as defined by the AMAGI Framework v2.0 Specification. The NIST AI RMF, published as NIST AI 100-1 (January 2023), organizes AI risk management activities into four functions: Govern, Map, Measure, and Manage. Each function contains subcategories that specify more granular risk management objectives. This profile maps all 17 subcategories to the AMAGI architectural mechanisms, providing suggested actions and implementation status for each subcategory.

This profile follows the methodology established by NIST AI 600-1 (Generative AI Profile) and the NIST CSF 2.0 Organizational Profile Template. Per NIST AI RMF Section 6, a profile is "an implementation of the AI RMF functions, categories, and subcategories for a specific setting or application." This profile applies the AI RMF to the AMAGI hardware-enforced safety architecture. Consistent with NIST methodology, this profile does not assign percentage-based coverage scores; instead, it provides an implementation status and suggested actions for each subcategory, allowing organizations to assess alignment with the AI RMF in the context of their specific risk management objectives and resources.

The AMAGI approach to AI risk management is fundamentally different from software-based governance frameworks. Where procedural controls rely on policies, training, and oversight that can be bypassed or degraded, AMAGI enforces risk management constraints through six hardware invariants that are physically impossible to violate. This structural approach eliminates entire categories of risk rather than merely mitigating them, providing a stronger and more reliable risk management posture than any procedural framework can achieve alone. The TTP (Trigger-Time-Delay-Policy-Activation) Triad serves as the foundational security primitive, ensuring that every critical operation requires explicit hardware approval before execution, with a mandatory time delay that provides human oversight opportunity.

This profile is intended for organizations seeking to claim alignment with the NIST AI RMF under the AI Safety Institute conformance program, regulatory bodies evaluating AI systems under the EU AI Act Article 9 (Risk Management System), and certification authorities assessing the AMAGI Architectural Class for Common Criteria evaluation. The profile complements the function-level mapping in Section 23 and the subcategory-level profile in Section 25 of the Framework v2.0 by providing a self-contained, publication-ready document that can be referenced independently.

## 2. Profile Methodology

This profile is structured according to the NIST AI RMF profile methodology, drawing on three authoritative sources for its format and approach. Understanding this methodology is essential for correctly interpreting the implementation status and suggested actions provided for each subcategory throughout this document.

### 2.1 Implementation Status (per NIST CSF 2.0)

Following the NIST CSF 2.0 Organizational Profile Template, each subcategory is assigned an implementation status that describes how the AMAGI architecture addresses that subcategory. The NIST CSF 2.0 template defines implementation statuses including Complete, Partial, Planned, Alternative, and Not Applicable. This profile adapts those categories to reflect the AMAGI architecture's unique characteristic of hardware-enforced risk management:

| Status                         | Definition                                                                                                                                                                                                                                                                                                 | NIST CSF 2.0 Equivalent                         |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Structural</b>              | The subcategory is fully addressed by hardware-enforced invariants. Compliance is mathematically guaranteed and cannot be bypassed through software vulnerabilities, configuration errors, or adversarial attacks.                                                                                         | Complete (implemented via structural control)   |
| <b>Structural + Procedural</b> | The subcategory is addressed by a combination of hardware-enforced guarantees and organizational processes. The hardware provides a structural foundation, while certain aspects (e.g., supply chain management, vendor selection) require procedural controls that cannot be fully automated in hardware. | Alternative (implemented via combined approach) |

Table 1: Implementation status definitions

## 2.2 Suggested Actions (per NIST AI 600-1)

Following the NIST AI 600-1 (Generative AI Profile) model, each subcategory includes suggested actions that describe specific AMAGI architectural mechanisms relevant to that subcategory. These suggested actions are not mandatory checklists but rather concrete mechanisms that organizations implementing the AMAGI Architectural Class can use to address the corresponding AI RMF subcategory. As NIST states in AI 600-1, suggested actions are "organized by relevant AI RMF subcategories" and provide operational guidance for managing risks within the profile's scope.

## 2.3 Profile Scope

This is a cross-sectoral target profile that applies the AI RMF to the AMAGI hardware-enforced safety architecture, independent of any specific deployment domain (medical, automotive, aerospace, industrial). The profile describes the target state of AI risk management for any system implementing the AMAGI Architectural Class. Organizations deploying AMAGI-based systems in specific sectors should use this profile as a foundation and extend it with sector-specific regulatory requirements (e.g., FDA PCCP for medical devices, ISO 26262 for automotive).

Per NIST AI RMF guidance, organizations are not expected to implement every subcategory with equal rigor; they should "document reasons for deferred work, along with compensating controls and implementation timelines." For the AMAGI Architectural Class, all subcategories are addressed either structurally or structurally with procedural overlay; none are deferred or excluded.

## 3. Scope and Normative References

This profile applies to the AMAGI Architectural Class as defined by the following normative specifications. All architectural claims, invariant definitions, and safety evidence referenced in this profile are authoritative only in the context of these specifications.

| Document                         | Version | Scope                                                                                  |
|----------------------------------|---------|----------------------------------------------------------------------------------------|
| AMAGI Framework Specification    | v2.0    | Protection Profile + Integration Specification; EAL2 augmented; SIL2; all 6 invariants |
| IEEE TCAD Paper (TCAD-2026-1011) | 2026    | Source of truth for numerical parameters (PFDavg, SFF, FIT, LUTs, latency)             |
| Amagi-TTP NIPU Specification     | v1.1    | Processor-level enforcement (SCC, MPE, ATL, EGW)                                       |
| Amagi-MEM Specification          | v1.1    | Memory-level enforcement (MAG, SHMF, audit diode)                                      |
| AMAGI Open Specification         | v1.1    | Public reference standard; regulatory and academic audience                            |
| NIST AI 100-1                    | 1.0     | AI Risk Management Framework (January 2023)                                            |
| NIST AI 600-1                    | 1.0     | Generative AI Profile (methodology reference for Suggested Actions)                    |
| ISO/IEC 15408                    | 2022    | Common Criteria for IT Security Evaluation                                             |
| IEC 61508                        | 2010    | Functional Safety of E/E/PE Safety-Related Systems                                     |

Table 2: Normative references

## 4. AMAGI Invariant Summary

The AMAGI Architectural Class is defined by six hardware-enforced invariants that collectively ensure that AI systems cannot bypass safety constraints, modify critical policies, or operate without complete audit accountability. These invariants form the structural foundation of the NIST AI RMF profile: each AI RMF subcategory is addressed through one or more invariants, providing a hardware-enforced guarantee rather than a procedural assurance. The invariants are organized in a three-layer dependency hierarchy: foundational (INV-001, INV-002), structural (INV-003, INV-004), and operational (INV-005, INV-006). This hierarchy ensures that the foundational invariants cannot be compromised without violating the structural and operational invariants as well, providing defense in depth.

| ID      | Invariant                                             | Layer        | NIST AI RMF Relevance                                                            |
|---------|-------------------------------------------------------|--------------|----------------------------------------------------------------------------------|
| INV-001 | Physical Non-Bypassability                            | Foundational | Ensures no governance bypass; structural enforcement of GV-1.2, MP-2.3           |
| INV-002 | Irreducible TCB (less than or equal to 500 lines HDL) | Foundational | Minimizes attack surface for MS-3.2; enables complete verification for ATE_COV.1 |
| INV-003 | Hardware Separation of Control and Cognition          | Structural   | Prevents unauthorized data access; enforces MP-2.2, FDP_ACC.1                    |

|         |                                            |             |                                                                      |
|---------|--------------------------------------------|-------------|----------------------------------------------------------------------|
| INV-004 | Controlled Mutability of Critical Policies | Structural  | Ensures policy integrity; enforces GV-1.1, GV-1.4, Dual-Tier Trust   |
| INV-005 | Complete Mediation                         | Operational | Guarantees all operations are mediated; enforces MS-3.1, MG-4.1      |
| INV-006 | Complete Audit Integrity                   | Operational | Provides tamper-proof risk evidence; enforces GV-1.7, MS-3.1, MS-3.3 |

Table 3: AMAGI invariant summary with NIST AI RMF relevance

## Part I: GOVERN (GV)

The GOVERN function establishes the organizational structures, policies, and accountability mechanisms for AI risk management. The AMAGI architecture implements governance through structural guarantees rather than procedural controls: the TTP Triad enforces governance as a hardware constraint, making violations physically impossible rather than merely policy-violating. This structural approach provides a stronger governance model than procedural governance alone, as it eliminates the possibility of governance bypass through software vulnerabilities, configuration errors, or adversarial attacks. The Dual-Tier Trust Model provides the governance framework for policy immutability, with Tier 1 representing the strongest governance (physically immutable policies via mask ROM and eFuse) and Tier 2 representing controlled governance (PQC-signed policy updates with hardware fallback to minimal-safe state). The Executive Gateway (EGW) enforces dual hardware approval for all actuator operations, ensuring that governance decisions are enforced at the hardware level.

### 5. GV-1.1: Documented AI Risk Management Policies

GV-1.1 requires that policies and procedures for AI risk management are documented and communicated across the organization. The AMAGI architecture satisfies this requirement through the Organizational Security Policies (OSPs) defined in Framework v2.0 Section 10, which specify five mandatory security policies (OSP.1 through OSP.5) covering separation of control and cognition, critical policy immutability, complete mediation, audit integrity, and post-market monitoring. These OSPs are not merely documented policies; they are enforced as hardware constraints by the six invariants, meaning that compliance is guaranteed by the architecture rather than relying on organizational discipline. The S-APL v2.0 license (Section 26) defines the governance framework for derivative architectures, ensuring that the documented policies apply to all implementations of the AMAGI Architectural Class. The ADR Register (Appendix H) provides the decision trail for all governance-related architectural decisions.

#### Suggested Actions:

- Implement OSP.1 through OSP.5 as the mandatory security policy baseline for all AMAGI-based systems
- Enforce OSP compliance through the six hardware invariants (INV-001 through INV-006), making policy violations architecturally impossible

- Use the S-APL v2.0 licensing framework to propagate documented policies to all derivative architectures
- Maintain the ADR Register for governance decision traceability

**Implementation Status:** Structural — OSP.1 through OSP.5 are hardware-enforced; policy documentation is structurally guaranteed by INV-004 (immutability) and INV-006 (audit trail).

**Evidence:** Framework v2.0 Sections 10, 26, Appendix H

## 6. GV-1.2: Accountability Structures

GV-1.2 requires that accountability structures for AI risk are established, with clearly defined roles and responsibilities. The AMAGI architecture satisfies this through the MAG policy matrix, which assigns per-cluster accountability for data access and operation authorization. The MAG (Memory Access Guard) is a 128x128 eFuse/OTP matrix that defines which cluster types can access which data regions and at which enforcement levels. This hardware-enforced accountability structure cannot be modified by software, ensuring that accountability assignments persist even under adversarial conditions. The TTP Triad enforces dual approval for critical operations: the SCC must approve the operation (TTP-approval) AND an external 2FA signal must be present (via the Executive Gateway). This dual approval structure provides clear accountability for every critical actuator operation, with the identity of the approving parties recorded in the tamper-proof audit trail (INV-006).

### Suggested Actions:

- Deploy the MAG policy matrix to assign per-cluster accountability for data access and operation authorization
- Configure the TTP Triad dual-approval mechanism (SCC TTP-approval + EGW 2FA) for all critical actuator operations
- Use INV-006 (complete audit integrity) to record accountability events in the tamper-proof audit trail

**Implementation Status:** Structural — accountability is enforced by MAG (eFuse/OTP) and TTP dual approval; bypass is physically impossible (INV-001).

**Evidence:** Framework v2.0 Sections 5, 14, 30; MEM v1.1 Section 3.2

## 7. GV-1.3: Workforce Diversity

GV-1.3 requires that workforce diversity and inclusion considerations are integrated into AI risk management. The AMAGI architecture addresses this through its Hard Macro delivery format, which enables diverse integration teams to work with the AMAGI IP Core without requiring access to the proprietary TCB internals. The RTL-Wrapper abstraction layer allows integration engineers from diverse backgrounds and skill levels to incorporate the AMAGI Hard Macro into their SoC designs through a well-defined interface, without needing to understand or modify the 500-line TCB implementation. This structural separation between the TCB (which requires deep security expertise) and the integration interface (which requires standard SoC design skills) ensures that a broader range of professionals can participate in AMAGI-based system development, supporting workforce diversity objectives.

### Suggested Actions:

- Use the Hard Macro delivery format (GDSII, SDC, RTL-Wrapper) to enable integration by diverse teams without TCB access
- Leverage the RTL-Wrapper abstraction to separate TCB expertise requirements from standard SoC design skills

**Implementation Status:** Structural — the Hard Macro format structurally separates TCB expertise from integration expertise; TCB modification is physically prevented (INV-002).

**Evidence:** Framework v2.0 Section 8.3

## 8. GV-1.4: Organizational Risk Tolerance

GV-1.4 requires that organizational tolerance for AI risk is explicitly defined. The AMAGI architecture provides a binary risk tolerance model through the Dual-Tier Trust Model (ADR-002), which defines two mutually exclusive operational regimes. Tier 1 (Hard Assurance) represents zero risk tolerance for policy modification: policies are stored in mask ROM and eFuse, which are physically immutable after fabrication. No software mechanism, configuration change, or administrative action can modify Tier 1 policies. Tier 2 (Controlled Assurance) represents controlled risk tolerance: policies can be updated through PQC-signed packages verified by Amagi-Q, with hardware fallback to the minimal-safe state if verification fails. This explicit, hardware-enforced binary model eliminates the ambiguity inherent in procedural risk tolerance definitions, where the boundary between acceptable and unacceptable risk is often subjective and inconsistently applied.

### Suggested Actions:

- Define organizational risk tolerance using the Dual-Tier Trust Model: Tier 1 (zero tolerance, physically immutable) or Tier 2 (controlled tolerance, PQC-signed updates)
- Configure PQC-signed policy update procedures for Tier 2 deployments with hardware fallback to minimal-safe state

**Implementation Status:** Structural — risk tolerance is hardware-encoded in mask ROM/eFuse (Tier 1) or enforced via PQC verification (Tier 2); modification is physically impossible (INV-004).

**Evidence:** Framework v2.0 Section 16

## 9. GV-1.5: Integrated Risk Management Practices

GV-1.5 requires that AI risk management practices are integrated into organizational processes and systems. The AMAGI architecture achieves this by design: all six invariants provide integrated risk management across all components of the TOE, from processor-level enforcement (SCC, MPE, ATL) to memory-level enforcement (MAG, SHMF) to actuator control (EGW) to boot-time verification (CRTM). The compositional verification (Framework v2.0 Section 32) formally proves that invariant preservation composes across component boundaries, ensuring that there are no gaps in the integrated risk management coverage. The assume-guarantee contracts for each component define exactly what risk management properties each component provides and what it requires from other components, creating a fully integrated risk management chain from power-on to actuator output.



**Suggested Actions:**

- Implement all six invariants (INV-001 through INV-006) across the full TOE component stack
- Verify invariant composition across component boundaries using assume-guarantee contracts (Section 32)
- Deploy the CRTM-to-EGW chain for end-to-end integrated risk management from power-on to actuator output

**Implementation Status:** Structural — all six invariants span the full TOE; compositional verification proves no gaps in risk management integration.

**Evidence:** Framework v2.0 Sections 4, 32

## 10. GV-1.6: Third-Party AI Risk Processes

GV-1.6 requires that processes for managing AI risks from third-party entities are defined. The AMAGI architecture addresses this through the Derivative Architecture Clause (Framework v2.0 Section 27), which defines the constraints on third-party modifications to AMAGI-based systems. The clause requires that all derivative architectures preserve the six invariants, pass the compliance verification requirements (Section 28), and operate under the S-APL v2.0 license. The compliance verification requirements define explicit acceptance criteria that third-party implementations must meet, including invariant preservation tests, FMEDA evidence requirements, and audit trail integrity verification. The C-SCRM (Cybersecurity Supply Chain Risk Management) measures per NIST SP 800-161 provide additional third-party risk controls for the supply chain, covering the fabrication, programming, and delivery stages where third-party involvement is highest.

While the AMAGI architecture provides strong structural protections against third-party risks (Hard Macro GDSII format makes post-fabrication TCB modification physically impossible, and compliance verification is hardware-testable), certain aspects of third-party risk management remain inherently procedural: vendor selection criteria, contract negotiation, delivery verification processes, and ongoing supply chain monitoring require organizational processes that cannot be fully encoded in hardware. The S-APL v2.0 license provides the legal enforcement framework that bridges the structural and procedural layers.

**Suggested Actions:**

- Apply the Derivative Architecture Clause (Section 27) to all third-party implementations, requiring invariant preservation and compliance verification
- Use the Hard Macro delivery format (GDSII) to structurally prevent post-fabrication TCB modification by third parties
- Implement C-SCRM controls per NIST SP 800-161 for fabrication, programming, and delivery stages
- Enforce S-APL v2.0 licensing for legal allocation of third-party risk responsibilities
- Establish organizational processes for vendor selection, contract negotiation, and delivery verification (procedural overlay)

**Implementation Status:** Structural + Procedural — TCB integrity is structurally guaranteed by Hard Macro format; supply chain management and vendor oversight require procedural overlay (C-SCRM per

NIST SP 800-161, S-APL v2.0 license enforcement).

**Evidence:** Framework v2.0 Sections 12, 27, 28

## 11. GV-1.7: Documented and Tracked Risk Practices

GV-1.7 requires that AI risk management practices are documented and tracked over time. The AMAGI architecture provides this through multiple complementary mechanisms. The ADR Register (Framework v2.0 Appendix H) tracks all architectural decisions that affect risk management, with eight ratified ADRs (ADR-001 through ADR-008) documenting the rationale and impact of each decision. The Change Log (Appendix D) tracks all modifications from v1.4 to v2.0, providing a complete history of risk management evolution. The hardware audit trail (INV-006) provides tamper-proof, cryptographically authenticated logging of all risk-relevant events during operation, including policy access, enforcement decisions, and actuator operations. The monotonic counters in the PQC key management system prevent replay attacks on the audit trail, ensuring that the tracking record cannot be forged or altered.

### Suggested Actions:

- Use the ADR Register for governance decision traceability and the Change Log for specification evolution tracking
- Deploy INV-006 (complete audit integrity) with hardware diode and PQC monotonic counters for tamper-proof operational tracking
- Configure FAU\_STG.3 CUTOFF to ensure audit storage does not silently fail (75% warning, 100% protected mode)

**Implementation Status:** Structural — all risk practices are tracked via hardware audit trail (INV-006); documentation integrity is guaranteed by tamper-proof logging with PQC counters.

**Evidence:** Framework v2.0 Sections 4, Appendix D, Appendix H

## 12. GOVERN Function Summary

| Subcategory | Primary Invariant       | Implementation Status   | Framework Section |
|-------------|-------------------------|-------------------------|-------------------|
| GV-1.1      | INV-004, INV-006        | Structural              | 10, 26, App H     |
| GV-1.2      | INV-001                 | Structural              | 5, 14, 30         |
| GV-1.3      | INV-002                 | Structural              | 8.3               |
| GV-1.4      | INV-004                 | Structural              | 16                |
| GV-1.5      | INV-001 through INV-006 | Structural              | 4, 32             |
| GV-1.6      | INV-004                 | Structural + Procedural | 12, 27, 28        |
| GV-1.7      | INV-006                 | Structural              | 4, App D, App H   |

Table 4: GOVERN function subcategory summary

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## Part II: MAP (MP)

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The MAP function identifies and contextualizes AI risks within the operational environment. The AMAGI architecture provides a comprehensive threat model under the "AI as Hostile Insider" assumption, which represents the most conservative risk framing possible for AI systems. Under this assumption, any AI behavior that violates the TTP predicate is treated as potentially hostile, regardless of intent. This framing eliminates the need for intent classification, which is notoriously unreliable for AI systems, and instead focuses on behavioral constraints enforced by hardware. The sensor trust chain (Framework v2.0 Section 37) extends the risk mapping to the input domain, addressing the often-overlooked risk of sensor spoofing and replay attacks that could cause an AI system to make decisions based on fabricated inputs.

### 13. MP-2.1: Contextual Understanding

MP-2.1 requires that the AI system is understood in its operational context, including the conditions under which it may produce erroneous or harmful outputs. The AMAGI architecture provides this through the TOE boundary definition (Framework v2.0 Section 9), which explicitly identifies all eight components within the security perimeter (CRTM, SCC, MPE, MAG, ATL, SHMF, Amagi-Q, EGW) and defines the trust boundaries between them. The cluster typing system (GEN, COM, ANL, EXE) contextualizes each AI function within the overall system architecture, defining which operations are permitted for each cluster type and at which enforcement levels. The threat model (Section 6) identifies six threat categories under the "AI as Hostile Insider" assumption, providing the contextual understanding of the threat landscape that the AI system operates within. The substrate independence claim (Section 7) ensures that the contextual understanding is portable across different hardware platforms.

#### Suggested Actions:

- Define the TOE boundary using the eight-component security perimeter and trust boundary model
- Apply the cluster typing system (GEN, COM, ANL, EXE) to contextualize AI functions and their permitted operations
- Use the "AI as Hostile Insider" threat model with six threat categories as the risk context baseline
- Leverage the substrate independence claim for cross-platform contextual portability

**Implementation Status:** Structural — TOE boundary and cluster typing are hardware-enforced; threat model is architecturally encoded in the TTP predicate.

**Evidence:** Framework v2.0 Sections 6, 7, 8, 9

### 14. MP-2.2: Training Data Understanding

MP-2.2 requires that the AI system is understood relative to its training and evaluation data, including data provenance and quality. The AMAGI architecture addresses this through the MAG policy matrix, which controls data access by cluster type, ensuring that training data provenance is tracked and that data access is limited to authorized clusters. The MPE (Memory Protection Engine) enforces data

governance at the memory level, ensuring that no data can be accessed without explicit policy authorization. The SHMF (Self-Healing Memory Framework) ensures data integrity through ECC and cryptographic hash chains (Phase 1) and AES-256/ML-KEM-1024 encryption (Phase 2), protecting the integrity of training data against corruption or tampering. The typed cluster boundaries (GEN, COM, ANL, EXE) prevent unauthorized cross-cluster data access, ensuring that training data is only accessible to authorized AI functions.

**Suggested Actions:**

- Deploy the MAG policy matrix to control data access by cluster type and track data provenance
- Use the MPE for memory-level data governance, ensuring no unauthorized data access
- Implement SHMF for data integrity (Phase 1: ECC + hash chains; Phase 2: AES-256/ML-KEM-1024 encryption)
- Enforce typed cluster boundaries to prevent unauthorized cross-cluster data access

**Implementation Status:** Structural — data access control is hardware-enforced by MAG and MPE; data integrity is guaranteed by SHMF; separation is enforced by INV-003.

**Evidence:** Framework v2.0 Sections 4, 14; MEM v1.1 Sections 3.1-3.4

## 15. MP-2.3: Intended Use Understanding

MP-2.3 requires that the AI system is understood relative to its intended use and the contexts in which it should not be used. The AMAGI architecture satisfies this through the OSPs (OSP.1 through OSP.5), which define the intended use constraints for the architectural class. The TTP Triad enforces operational boundaries by requiring explicit approval for every critical operation, with the approval criteria defined by the MAG policy matrix. The EGW ensures that actuator operations remain within scope by requiring dual hardware approval (SCC TTP-approval AND external 2FA), preventing the AI system from performing operations outside its intended use. The graceful degradation policy (Section 33) defines the system behavior when operational boundaries are approached or exceeded, ensuring that the system transitions to a safe state rather than continuing to operate outside its intended use.

**Suggested Actions:**

- Define intended use constraints through OSP.1 through OSP.5 and enforce them via hardware invariants
- Configure TTP Triad approval criteria via the MAG policy matrix for operational boundary enforcement
- Deploy EGW dual approval (SCC + 2FA) to prevent out-of-scope actuator operations
- Implement graceful degradation policy (Section 33) for safe-state transitions when boundaries are exceeded

**Implementation Status:** Structural — intended use boundaries are enforced by TTP Triad and EGW; out-of-scope operations are physically impossible (INV-001).

**Evidence:** Framework v2.0 Sections 5, 10, 30, 33

## 16. MP-2.4: Deployment Context Understanding

MP-2.4 requires that the AI system is understood relative to its deployment context, including the operational environment and regulatory requirements. The AMAGI architecture addresses this through the substrate independence claim (Section 7), which ensures that the security architecture is portable across different hardware platforms, with a reference target of Xilinx Artix-7 FPGA. The CRTM (Section 29) verifies the deployment integrity at power-on reset, ensuring that the enforcement mechanism is intact before the system becomes operational in any deployment context. The Dual-Tier Trust Model (Section 16) addresses the regulatory context by providing two compliance tiers: Tier 1 for high-regulation environments (medical, automotive, aerospace) and Tier 2 for environments where controlled policy updates are needed (e.g., FDA PCCP compliance). The EU AI Act Annex IV mapping (Section 22) and this NIST AI RMF profile provide the regulatory context documentation required for deployment in regulated industries.

**Suggested Actions:**

- Verify deployment integrity using CRTM at power-on reset before system becomes operational
- Select appropriate Dual-Tier Trust level for the deployment context (Tier 1: high-regulation; Tier 2: controlled updates)
- Use the substrate independence claim for cross-platform deployment portability (reference: Artix-7 FPGA)
- Apply EU AI Act Annex IV mapping and this NIST AI RMF profile for regulatory context documentation

**Implementation Status:** Structural + Descriptive — deployment integrity is verified by CRTM; compliance tiers are hardware-encoded; regulatory mappings are documented through descriptive Annex IV and NIST profile materials.

**Evidence:** Framework v2.0 Sections 7, 16, 22, 29

## 17. MP-2.5: Impact Understanding

MP-2.5 requires that the AI system is understood relative to its potential impacts, including both beneficial and harmful impacts. The AMAGI architecture provides this through the threat model (Section 6), which identifies six threat categories and maps them to security objectives and SFRs. The graceful degradation hierarchy (Section 33.3) prioritizes safety-critical functions by defining which invariants must be preserved at each degradation level, providing a clear impact hierarchy. The FMEDA (Section 17) quantifies the failure impact in FIT (Failures In Time), providing quantitative evidence that dangerous undetected failures are below the SIL2 threshold ( $SFF = 93\%$ ,  $PFD_{avg} = 6.9 \times 10^{-5}$ ). The medical device safe state requirements (Section 33.4) define the impact boundaries for the highest-risk deployment context, ensuring that the system always transitions to a safe state rather than producing uncontrolled outputs.

**Suggested Actions:**

- Use the threat model (6 categories) to identify and map potential impacts to security objectives
- Deploy the graceful degradation hierarchy to prioritize invariant preservation by impact severity

- Apply FMEDA quantitative analysis (SFF = 93%, PFDavg =  $6.9 \times 10^{-5}$ ) for evidence-based impact assessment
- Implement medical device safe state requirements for the highest-risk deployment contexts

**Implementation Status:** Structural + Descriptive — impact boundaries are enforced by INV-005 (complete mediation) and graceful degradation; quantitative impact evidence provided by FMEDA; safe-state descriptions for medical devices provided as procedural overlay.

**Evidence:** Framework v2.0 Sections 6, 17, 33; TCAD-2026-1011

## 18. MAP Function Summary

| Subcategory | Primary Invariant | Implementation Status | Framework Section |
|-------------|-------------------|-----------------------|-------------------|
| MP-2.1      | INV-001           | Structural            | 6, 7, 8, 9        |
| MP-2.2      | INV-003           | Structural            | 4, 14             |
| MP-2.3      | INV-001           | Structural            | 5, 10, 30, 33     |
| MP-2.4      | ADR-002, ADR-007  | Structural            | 7, 16, 22, 29     |
| MP-2.5      | INV-005           | Structural            | 6, 17, 33         |

Table 5: MAP function subcategory summary

## Part III: MEASURE (MS)

The MEASURE function establishes quantitative and qualitative methods for assessing AI risks. The AMAGI architecture provides multiple measurement mechanisms spanning formal verification, quantitative safety analysis, and runtime monitoring. The SPIN model checker provides exhaustive state-space measurement of the SCC enforcement logic, verifying 15 LTL properties across 25,927,680 states: 9 safety properties unconditionally pass (zero violations); 2 architectural assumptions are trivially satisfied; and 4 liveness properties (SCC-L01 through SCC-L03) require fairness assumptions (FA1, FA2) to resolve counterexamples, while SCC-L04 remains open under the current model. The FMEDA provides quantitative failure rate measurement ( $\lambda_{DU} = 18.9 \text{ FIT}$ , SFF = 93%, PFDavg =  $6.9 \times 10^{-5}$ ) supporting the SIL2 safety claim. The runtime attestation mechanisms provide continuous measurement of system health through periodic self-tests (60s to 30min intervals), challenge-response attestation, and continuous audit-based monitoring. Together, these measurement mechanisms provide a comprehensive, multi-layered risk measurement framework that addresses both static (design-time) and dynamic (runtime) risk assessment.

### 19. MS-3.1: AI Risk Assessment and Tracking

MS-3.1 requires that AI risks are assessed and tracked using established methods. The AMAGI architecture provides three complementary risk assessment methods. First, the SPIN model checker assesses all reachable SCC enforcement states through exhaustive state-space exploration, providing a

mathematically guaranteed risk assessment for the TTP logic. Second, the FMEDA tracks hardware failure rates using the IEC 61508 methodology, providing quantitative risk assessment in FIT (Failures In Time) with diagnostic coverage analysis. Third, the hardware audit trail (INV-006) records all risk-relevant events during operation, providing a tamper-proof tracking record that can be analyzed for risk trends. The SFR-Invariant-Test traceability matrix (Section 35) provides the assessment coverage analysis, mapping all 18 SFRs through 6 invariants to 24 concrete test cases. The runtime attestation mechanisms (Section 34) provide ongoing risk tracking through periodic self-tests and continuous monitoring.

**Suggested Actions:**

- Use SPIN model checking for exhaustive state-space risk assessment of the SCC (25,927,680 states, 15 LTL properties: 9 safety unconditionally pass, 4 liveness conditional on fairness assumptions)
- Apply FMEDA (IEC 61508 methodology) for quantitative failure rate tracking ( $\lambda_{DU} = 18.9$  FIT)
- Deploy INV-006 audit trail for tamper-proof operational risk tracking
- Implement SFR-Invariant-Test traceability matrix (18 SFRs, 6 invariants, 24 test cases) for assessment coverage
- Configure runtime attestation (60s-30min self-tests, challenge-response) for ongoing risk tracking

**Implementation Status:** Structural — risk assessment is formally verified (SPIN), quantitatively tracked (FMEDA), and continuously monitored (INV-006 audit trail + runtime attestation).

**Evidence:** Framework v2.0 Sections 17, 20, 34, 35; TCAD-2026-1011

## 20. MS-3.2: Reliable Risk Assessment Tools

MS-3.2 requires that AI risk assessment tools and metrics are reliable and validated. The AMAGI architecture provides three validated risk assessment tools. The formal verification via SPIN provides mathematically guaranteed assessment results: the model checker exhaustively explores all reachable states and verifies LTL properties with zero false positives or false negatives. The FMEDA follows the IEC 61508-6 methodology, which is the internationally recognized standard for quantitative safety assessment, providing validated failure rate data with diagnostic coverage analysis. The runtime attestation provides independent verification through challenge-response protocols with nonce-based replay protection, ensuring that assessment results are fresh and authentic. The compositional verification (Section 32) provides validated composition of individual component assessments into a system-level assessment through assume-guarantee contracts.

**Suggested Actions:**

- Use SPIN model checker for formally verified risk assessment with zero false positives/negatives
- Apply IEC 61508-6 FMEDA methodology for internationally validated quantitative safety assessment
- Deploy challenge-response attestation with nonce-based replay protection for independent verification
- Leverage compositional verification (assume-guarantee contracts) for validated system-level assessment

**Implementation Status:** Structural — all risk assessment tools are formally validated or internationally standardized; assessment integrity is guaranteed by INV-006.

**Evidence:** Framework v2.0 Sections 20, 32, 34, 36

## 21. MS-3.3: Documented Risk Assessment Results

MS-3.3 requires that AI risk assessment results are documented and maintained. The AMAGI architecture provides this through multiple documentation mechanisms. The SPIN model checking results are documented in Framework v2.0 Section 20, including the number of states explored, the LTL properties verified, and the verification outcome (zero violations). The FMEDA results are documented in Section 17 and Appendix B, with detailed component-level failure mode analysis including failure rates, diagnostic coverage, and safe failure fraction calculations. The updated FMEDA for CRTM and EGW is documented in Section 17 (FMEDA Summary) and Appendix B (FMEDA Component-Level Breakdown). The runtime attestation results are logged to the hardware audit trail (INV-006), which provides tamper-proof documentation of all assessment events. The FAU\_STG.3 CUTOFF mechanism ensures that the audit storage does not silently fail, alerting operators when storage reaches 75% capacity and entering a protected mode at 100% capacity that denies all access until the audit data is offloaded.

### Suggested Actions:

- Document SPIN verification results (25,927,680 states, 15 LTL properties: 9 safety zero violations, 4 liveness conditional on fairness FA1/FA2) per Section 20
- Maintain FMEDA documentation (component-level failure rates, diagnostic coverage, SFF calculations) per Section 17 and Appendix B
- Use INV-006 hardware audit trail with FAU\_STG.3 CUTOFF for tamper-proof, fail-safe documentation of assessment results

**Implementation Status:** Structural — assessment documentation integrity is guaranteed by INV-006 audit trail; storage reliability is ensured by FAU\_STG.3 CUTOFF mechanism.

**Evidence:** Framework v2.0 Sections 17, 20, 34, 38

## 22. MEASURE Function Summary

| Subcategory | Primary Invariant | Implementation Status | Framework Section |
|-------------|-------------------|-----------------------|-------------------|
| MS-3.1      | INV-006           | Structural            | 17, 20, 34, 35    |
| MS-3.2      | ADV_SPM.1         | Structural            | 20, 32, 34, 36    |
| MS-3.3      | FAU_STG.3         | Structural            | 17, 20, 34, 38    |

Table 6: MEASURE function subcategory summary

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## Part IV: MANAGE (MG)

The MANAGE function implements risk treatment strategies and monitors their effectiveness over time. The AMAGI architecture provides structural risk treatment through the invariant enforcement model:



each invariant is a hardware-enforced constraint that eliminates an entire category of risk rather than merely mitigating it. The graceful degradation policy (Section 33) provides the risk treatment strategy for component failures, ensuring that the system transitions to a safe state while preserving as many invariants as possible. The Dual-Tier Trust Model provides the risk treatment framework for policy modification, with Tier 1 representing risk elimination (physically immutable policies) and Tier 2 representing risk mitigation (controlled policy updates with hardware fallback). The runtime attestation (Section 34) provides the ongoing monitoring mechanism that verifies the effectiveness of risk treatments in real-time, following the NIST SP 800-193 PROTECT/DETECT/RECOVER framework.

## 23. MG-4.1: Risk Treatment Based on Assessment

MG-4.1 requires that AI risks are treated based on the results of risk assessments. The AMAGI architecture treats risks through three complementary mechanisms. First, the six invariants structurally eliminate risk categories: INV-001 eliminates bypass risk, INV-002 eliminates TCB complexity risk, INV-003 eliminates data access risk, INV-004 eliminates policy modification risk, INV-005 eliminates unmediated operation risk, and INV-006 eliminates audit tampering risk. Second, the graceful degradation policy treats the residual risk of component failures by defining seven degradation modes that preserve the maximum number of invariants possible for each failure scenario, with the system always transitioning toward a safer state. Third, the sensor trust chain (Section 37) treats the input-side risk by defining three sensor trust levels (Authenticated, Authenticated+Redundant, Authenticated+Redundant+Attested) that provide increasing levels of input integrity assurance.

### Suggested Actions:

- Implement all six invariants for structural elimination of risk categories (bypass, complexity, data access, policy modification, unmediated operation, audit tampering)
- Deploy graceful degradation policy (7 modes) for residual risk treatment of component failures
- Configure sensor trust chain (3 levels) for input-side risk treatment

**Implementation Status:** Structural — risk treatment is hardware-enforced through invariants; residual risk is managed through graceful degradation and sensor trust chain.

**Evidence:** Framework v2.0 Sections 4, 33, 37

## 24. MG-4.2: Prioritized Risk Treatments

MG-4.2 requires that AI risk treatments are prioritized based on assessed impact and likelihood. The AMAGI architecture provides prioritization through the invariant dependency graph (Section 31), which defines a three-layer priority hierarchy. Foundational invariants (INV-001, INV-002) have the highest priority because all other invariants depend on them: if either INV-001 or INV-002 is violated, none of the structural or operational invariants can hold. Structural invariants (INV-003, INV-004) have the second priority because they depend on foundational invariants and support operational invariants. Operational invariants (INV-005, INV-006) have the third priority. The graceful degradation hierarchy respects this priority: the system always preserves foundational invariants first, then structural, then operational, ensuring that the most critical risk treatments are maintained even under component failure. The cascade failure analysis (Section 31.5) demonstrates that no single component failure can cause a

foundational invariant violation without first degrading through the structural and operational layers.

**Suggested Actions:**

- Follow the invariant dependency graph (foundational > structural > operational) for risk treatment prioritization
- Configure graceful degradation to respect the three-layer priority hierarchy
- Verify cascade failure analysis to ensure no single component failure causes foundational invariant violation

**Implementation Status:** Structural — risk treatment prioritization is encoded in the invariant dependency graph and enforced by graceful degradation.

**Evidence:** Framework v2.0 Sections 31, 33

## 25. MG-4.3: Third-Party Risk Management

MG-4.3 requires that AI risks from third-party entities are managed throughout the AI system lifecycle. The AMAGI architecture provides this through the Derivative Architecture Clause (Section 27), which constrains third-party modifications by requiring that all derivative architectures preserve the six invariants. The compliance verification requirements (Section 28) define explicit acceptance criteria that third-party implementations must meet, including invariant preservation tests, FMEDA evidence requirements, and audit trail integrity verification. The C-SCRM (Cybersecurity Supply Chain Risk Management) measures per NIST SP 800-161 provide additional third-party risk controls for the supply chain, covering the fabrication, programming, and delivery stages. The Hard Macro delivery format (GDSII, SDC, RTL-Wrapper) structurally prevents third-party modification of the TCB, as the GDSII mask-level representation cannot be altered without a full mask set change at the foundry. The S-APL v2.0 license provides the legal framework for third-party risk allocation and enforcement.

As with GV-1.6, certain aspects of third-party risk management remain inherently procedural: ongoing supply chain monitoring, vendor compliance auditing, and contractual enforcement of security requirements across the supply chain lifecycle require organizational processes that supplement the structural guarantees. The Hard Macro format provides the structural foundation (TCB integrity is physically protected), while C-SCRM and S-APL v2.0 provide the procedural framework for managing third-party risks that cannot be fully automated in hardware.

**Suggested Actions:**

- Apply the Derivative Architecture Clause (Section 27) to all third-party implementations
- Use Hard Macro delivery format (GDSII) to structurally prevent post-fabrication TCB modification
- Implement compliance verification requirements (Section 28) for invariant preservation tests and FMEDA evidence
- Deploy C-SCRM controls per NIST SP 800-161 for supply chain risk management (procedural overlay)
- Enforce S-APL v2.0 licensing for legal framework of third-party risk allocation (procedural overlay)

**Implementation Status:** Structural + Procedural — TCB integrity is structurally guaranteed by Hard Macro format; supply chain lifecycle management requires procedural overlay (C-SCRM, S-APL v2.0).

**Evidence:** Framework v2.0 Sections 12, 27, 28

## 26. MANAGE Function Summary

| Subcategory | Primary Invariant       | Implementation Status   | Framework Section |
|-------------|-------------------------|-------------------------|-------------------|
| MG-4.1      | INV-001 through INV-006 | Structural              | 4, 33, 37         |
| MG-4.2      | INV-001, INV-002        | Structural              | 31, 33            |
| MG-4.3      | INV-004                 | Structural + Procedural | 12, 27, 28        |

Table 7: MANAGE function subcategory summary

## 27. Comprehensive Subcategory Summary

The following table provides a comprehensive view of all 17 NIST AI RMF subcategories mapped to the AMAGI Architectural Class, with implementation status and suggested actions detailed in the preceding sections. Consistent with NIST profile methodology, this table uses implementation status rather than percentage-based coverage metrics. All subcategories are addressed by the AMAGI architecture: 15 subcategories are addressed structurally through hardware-enforced invariants, and 2 subcategories (GV-1.6, MG-4.3) require a structural + procedural approach due to the inherently organizational nature of supply chain risk management.

| ID     | Subcategory                | Invariant(s)        | Implementation Status   | FWK Section |
|--------|----------------------------|---------------------|-------------------------|-------------|
| GV-1.1 | Documented policies        | INV-004, INV-006    | Structural              | 10, 26      |
| GV-1.2 | Accountability structures  | INV-001             | Structural              | 5, 14, 30   |
| GV-1.3 | Workforce diversity        | INV-002             | Structural              | 8.3         |
| GV-1.4 | Risk tolerance             | INV-004             | Structural              | 16          |
| GV-1.5 | Integrated practices       | INV-001 through 006 | Structural              | 4, 32       |
| GV-1.6 | Third-party risk processes | INV-004             | Structural + Procedural | 12, 27, 28  |
| GV-1.7 | Tracked practices          | INV-006             | Structural              | 4, App D/H  |
| MP-2.1 | Contextual understanding   | INV-001             | Structural              | 6, 7, 8, 9  |
| MP-2.2 | Training data              | INV-003             | Structural              | 4, 14       |

|        |                        |                     |                                 |                |
|--------|------------------------|---------------------|---------------------------------|----------------|
| MP-2.3 | Intended use           | INV-001             | <b>Structural</b>               | 5, 10, 30, 33  |
| MP-2.4 | Deployment context     | ADR-002, ADR-007    | <b>Structural + Descriptive</b> | 7, 16, 22, 29  |
| MP-2.5 | Impact understanding   | INV-005             | <b>Structural + Descriptive</b> | 6, 17, 33      |
| MS-3.1 | Risk assessment        | INV-006             | <b>Structural</b>               | 17, 20, 34, 35 |
| MS-3.2 | Reliable tools         | ADV_SPM.1           | <b>Structural</b>               | 20, 32, 34, 36 |
| MS-3.3 | Documented results     | FAU_STG.3           | <b>Structural</b>               | 17, 20, 34, 38 |
| MG-4.1 | Risk treatment         | INV-001 through 006 | <b>Structural</b>               | 4, 33, 37      |
| MG-4.2 | Prioritized treatment  | INV-001, INV-002    | <b>Structural</b>               | 31, 33         |
| MG-4.3 | Third-party management | INV-004             | <b>Structural + Procedural</b>  | 12, 27, 28     |

Table 8: Comprehensive NIST AI RMF subcategory-to-AMAGI mapping

## 28. Implementation Approach Summary

This section summarizes the implementation approach across all 17 NIST AI RMF subcategories, consistent with the NIST profile methodology. Rather than using percentage-based coverage metrics (which are not part of NIST methodology), this analysis describes the dominant implementation approach and its implications for risk management assurance. The AMAGI architecture's distinguishing characteristic is that the vast majority of subcategories are addressed through structural (hardware-enforced) controls, in contrast to software-only governance frameworks where all controls are procedural.

| Implementation Approach         | Subcategories                                                                               | Count | Assurance Characteristics                                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structural</b>               | GV-1.1 through GV-1.5, GV-1.7, MP-2.1 through MP-2.3, MS-3.1 through MS-3.3, MG-4.1, MG-4.2 | 13    | Hardware-enforced by invariants; mathematically guaranteed; cannot be bypassed through software vulnerabilities, configuration errors, or adversarial attacks |
| <b>Structural + Descriptive</b> | MP-2.4, MP-2.5                                                                              | 2     | Hardware-enforced core with descriptive regulatory documentation overlay (EU AI Act Annex IV, NIST profile, medical device safe-state descriptions)           |

|                                |                |   |                                                                                                                                                                                               |
|--------------------------------|----------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structural + Procedural</b> | GV-1.6, MG-4.3 | 2 | TCB integrity is structurally guaranteed by Hard Macro (GDSII) format; supply chain lifecycle management requires organizational processes (C-SCRM per NIST SP 800-161, S-APL v2.0 licensing) |
|--------------------------------|----------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table 9: Implementation approach distribution

The dominance of structural implementation is a significant differentiator of the AMAGI Architectural Class compared to software-only AI governance frameworks. In a software-only framework, all 17 subcategories would be implemented procedurally, relying on policies, training, and oversight that can be bypassed or degraded through software vulnerabilities, configuration errors, or adversarial attacks. The AMAGI architecture's hardware-enforced approach eliminates these bypass vectors for 13 of 17 subcategories, with 2 additional subcategories having a structural core reinforced by descriptive documentation, providing a stronger assurance model than any procedural framework can achieve alone. The two subcategories requiring a structural + procedural approach (GV-1.6 and MG-4.3) address supply chain risks that involve human processes such as vendor selection, contract negotiation, delivery verification, and ongoing compliance monitoring. These processes are inherently organizational and cannot be fully automated in hardware. However, even these procedural controls are anchored by structural guarantees: the Hard Macro delivery format (GDSII) makes post-fabrication TCB modification physically impossible, and the S-APL v2.0 license provides legal enforcement of the procedural requirements. This means that while the procedural overlay must be managed by the organization, the structural foundation ensures that the most critical security properties (TCB integrity, policy immutability) cannot be compromised regardless of procedural failures.

## 29. Version History

| Version | Date     | Changes                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | Dec 2025 | Initial release. Subcategory-level profile mapping all 17 NIST AI RMF subcategories to AMAGI invariants, SFRs, and architectural mechanisms. Cross-references Framework v2.0 (Sections 23, 40).                                                                                                                                                                                                                   |
| 1.1     | Jun 2026 | Rewritten following NIST AI RMF profile methodology (AI 600-1 Suggested Actions model, CSF 2.0 Implementation Status model). Replaced percentage-based coverage metrics with implementation status and suggested actions. Corrected PFDavg to $6.9 \times 10^{-5}$ per TCAD-2026-1011. Added TCAD paper as normative reference. Added Profile Methodology section. Added EU AI Act regulatory context for MP-2.4. |

Table 10: Version history